

ViT + XAI on 90 × 65 Technical-Indicator Images

Workflow plan — NIFTY 50 Buy / Hold / Sell prediction

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Base paper: Gezici & Sefer, “Deep Transformer-Based Asset Price and Direction Prediction,” IEEE Access, vol. 12, 2024.

Setup

Item	Choice
Input image	90 indicators (rows) × 65 trading days (columns), 1 channel — shape (1, 90, 65)
Days	65 — same window as the base paper; $65 = 13 \times 5$, so it divides evenly by the patch width
Patch size	$10 \times 13 \rightarrow 90 \div 10 = 9$ rows of patches, $65 \div 13 = 5$ columns $\rightarrow 9 \times 5 = 45$ tokens
Model	Vision Transformer (ViT)
Explainability	XAI on the trained ViT to show which indicators and days drive each decision
Data	NIFTY 50 stocks, daily OHLCV
Labels	Buy / Hold / Sell from next-day return, threshold 0.01 (as in the base paper)
Train / test	5-year training window, next 1 year as test, slide forward 1 year (as in the base paper)

Patch check: $10 \times 9 = 90$ and $13 \times 5 = 65$, so the image is covered exactly with no cropping or padding. Each patch covers 10 indicators over 13 days.

Phases at a glance

Phase	Name	Duration
1	Data collection	1 week
2	90 indicators	1 week
3	Labels and train/test splits	0.5 week
4	90 × 65 image generation	0.5 week
5	ViT training	2 weeks
6	Evaluation	1 week
7	XAI	2 weeks
8	Write-up	2 weeks

Phase 1 — Data collection (1 week)

1. Download daily OHLCV for NIFTY 50 stocks (.NS tickers) from yfinance, for example 2008–2025.
2. Keep stocks with full price history over the period.
3. Remove duplicate dates and fix or drop missing days.
4. Save the cleaned data per stock.

Output: clean daily price data for each stock.

Phase 2 — 90 indicators (1 week)

1. Compute 90 indicators with TA-Lib: the base paper's 65 plus 25 more (for example ATR, NATR, and extra periods such as RSI 7/21). The list is in the appendix.
2. Group rows by category (overlap, momentum, volume, volatility, price transform, statistics) and keep this order fixed.
3. Drop the first rows of each stock where indicators are still empty (warm-up).

Output: a table of 90 indicator values per day per stock.

Phase 3 — Labels and train/test splits (0.5 week)

1. Label each day from the next day's return:

```
r = Close(t+1) / Close(t) - 1  
Buy if r > 0.01 | Sell if r < -0.01 | Hold otherwise
```

2. Split by time: train on 5 years, test on the next year, then slide forward 1 year (e.g. train 2008–2012 → test 2013, train 2009–2013 → test 2014).

Output: labels for every day and the list of train/test year ranges.

Phase 4 — 90 × 65 image generation (0.5 week)

1. For each day t , take the last 65 days ($t-64$ to t) of all 90 indicators.
2. Arrange as rows = indicators, columns = days (oldest on the left) → a 90×65 array.
3. Standardize each indicator using mean and standard deviation from the training years only, then apply the same values to the test year.
4. Plot a few Buy, Hold, and Sell images to check they look sensible.

Output: (1, 90, 65) image arrays with labels, ready for training.

Phase 5 — ViT training (2 weeks)

1. ViT configuration:

```
img_size=(90, 65), in_chans=1, num_classes=3  
patch_size=(10, 13) -> 9 x 5 = 45 tokens  
embed_dim=192, depth=6, num_heads=6
```

2. Train on each training window with cross-entropy loss and Adam/AdamW; stop when validation loss stops improving.
3. Save the trained model and its test-year predictions for each window.

Output: trained ViT models and test predictions.

Phase 6 — Evaluation (1 week)

1. Compute accuracy, precision, recall, and F1 for Buy, Hold, and Sell from the saved predictions.
2. Build the confusion matrix and check that accuracy equals the diagonal total divided by all samples (this is where the base paper's numbers did not match).
3. Summarize results per test year and overall.

Output: results tables and confusion matrices.

Phase 7 — XAI (2 weeks)

1. Apply XAI to the trained ViT, for example attention rollout and Integrated Gradients.
2. Each explanation gives an importance score for each of the 45 patches; map it back onto the 90 × 65 grid.
3. Sum importance across columns to rank the 90 indicators, and across rows to see which recent days matter most.
4. Compare what drives Buy versus Sell versus Hold predictions.
5. Make heatmaps with indicator names on the rows and days on the columns.

Output: indicator rankings per class and XAI heatmaps.

Phase 8 — Write-up (2 weeks)

1. Sections: introduction, related work, data and 90 × 65 image encoding, ViT model, results, XAI findings, conclusion.
2. All tables and figures taken directly from Phases 6 and 7.

Output: thesis chapter and paper draft.

Appendix — 90 indicators by category

Category	Count	Indicators
Overlap studies	18	SMA 10/20/50, EMA 10/20/50, WMA 20, DEMA 20, TEMA 20, TRIMA 20, KAMA 20, MIDPOINT 14, MIDPRICE 14, BBANDS upper/middle/lower 20, HT_TRENDLINE, SAR
Momentum	38	RSI 7/14/21, ROC 5/10/20, MOM 10/20, CMO 14, WILLR 14/28, CCI 14/20, MACD / signal / histogram, PPO, APO, ADX 14, ADXR, DX, PLUS_DI, MINUS_DI, AROON up/down, AROONOSC, STOCH K/D, STOCHF K/D, STOCHRSI K/D, MFI 14, TRIX 15, ULTOSC, BOP
Volume	6	OBV, AD, ADOSC 3/10, volume ÷ 20-day average volume, MFI 28
Volatility	8	ATR 14/28, NATR 14/28, TRANGE, return std 10/20, BBANDS width 20
Price transform	4	AVGPRICE, MEDPRICE, TYPPRICE, WCLPRICE
Statistics	16	BETA 20, CORREL 20, LINEARREG, LINEARREG_ANGLE, LINEARREG_SLOPE, LINEARREG_INTERCEPT, TSF, STDDEV, VAR, HT_DCPERIOD, HT_DCPHASE, HT_PHASOR (2), HT_SINE (2), HT_TRENDMODE
Total	90	